Oligopoly

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 18, Oligopoly.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 18, "Oligopoly".

What you must know from this chapter

1. **Conditions:** what makes a market an oligopoly.
2. **Cartels:** firms acting together as one monopoly.
3. **Collusion:** agreements to restrict output or fix prices, and why they break down.
4. How the outcome under oligopoly compares with monopoly and perfect competition.

The model used in this lecture

Two firms, A and B, sell an identical product.

- Market demand is a straight line:

$$P = a - bQ, \qquad Q = q_A + q_B.$$

- q_A and q_B are the two firms' outputs. Q is total output.
- Both firms have the same constant marginal cost, c, and no fixed cost.
- So each unit costs c, and average cost is also c.

Part 1

What Makes a Market an Oligopoly?

Oligopoly

Definition: Oligopoly
A market with only a few sellers, each offering a product similar or identical to the others'.

- **Few sellers:** a handful of firms supply most of the market.
- **Similar or identical products.**
- **Barriers to entry:** new firms find it hard to enter.
- **Interdependence:** each firm's profit depends on what its rivals do.

Examples: commercial aircraft (Boeing and Airbus); U.S. wireless phone service.

Definition: Duopoly
An oligopoly with exactly two sellers.

Interdependence, written as an equation

Firm A's profit is its profit per unit times its output.

$$
\begin{aligned}
\pi_A &= (P - c) \times q_A && \text{profit per unit times units} \\
&= (a - bQ - c) \times q_A && \text{replace } P \text{ with } a - bQ \\
&= (a - b\,q_A - b\,q_B - c) \times q_A && \text{replace } Q \text{ with } q_A + q_B
\end{aligned}
$$

- Firm B's output, q_B, appears in firm A's profit.
- The more B sells, the lower the price, and the less A earns on each unit.

Finding

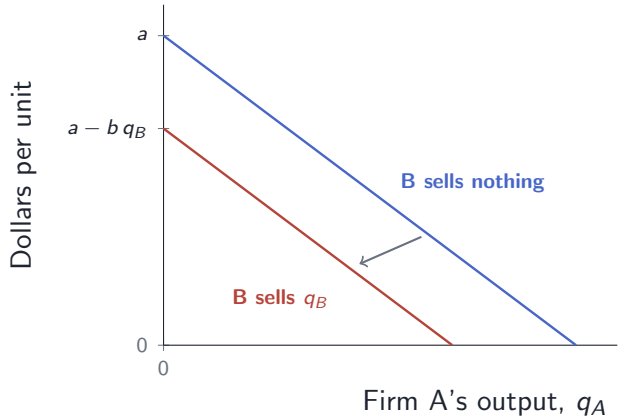
In an oligopoly, a firm's profit depends on its rival's choice. In perfect competition and in monopoly, no rival's choice enters a firm's profit.

The demand left for firm A

- Rearrange demand:

$$P = (a - b\,q_B) - b\,q_A.$$

- For a given q_B, this is the demand left for firm A.
- It starts at $a - b\,q_B$ and has slope $-b$.
- When B sells more, A's demand shifts down.



Four market structures

	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Sellers	Many	Many	**A few**	One
Product	Identical	Differentiated	**Similar or identical**	No close substitutes
Entry	Free	Free	**Difficult**	Blocked
Rivals matter?	No	Little	**Yes**	No rivals

Finding

What sets oligopoly apart is interdependence: each firm must allow for what its rivals do.

Part 1 summary

- Oligopoly: a few sellers of similar or identical products.
- Entry is difficult.
- Interdependence: firm A's profit depends on firm B's output.
- When B sells more, the demand left for A shifts down.
- Compared with perfect competition and monopoly: there, no rival's choice enters a firm's profit.

Check: Part 1

Which of the following examples illustrates an oligopoly market?

A A farmers' market with many individuals selling sweet corn and tomatoes

B A city whose electrical service is provided by one electric co-operative

C A city with two firms that are licensed to sell school uniforms for the local schools

D A city with many independently owned hair styling salons

Check: Part 1 – answer

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Finding

Two sellers of the same product, with entry blocked by licensing: a duopoly. A is perfect competition, B is monopoly, D is monopolistic competition.

Part 2

Cartels: Acting as One Monopoly

Collusion and cartels

Definition: Collusion

An agreement among firms in a market about the quantities to produce or the prices to charge.

Definition: Cartel

A group of firms acting together. A successful cartel acts as one monopolist.

- The cartel chooses total output Q to maximize joint profit.
- Then it splits Q between the members.

The cartel's output: the monopoly rule

As one monopolist, the cartel produces where MR = MC.

$\text{MR} = a - 2bQ$	straight-line demand: MR twice as steep (Chapter 16)
$a - 2bQ = c$	set MR = MC
$2bQ = a - c$	add $2bQ$ to both sides; subtract c
$Q_{\text{cartel}} = \dfrac{a-c}{2b}$	divide both sides by $2b$

Price, from demand:

$$P_{\text{cartel}} = a - b \times \frac{a-c}{2b}$$ put Q_{cartel} into $P = a - bQ$

$$= a - \frac{a-c}{2} = \frac{a+c}{2}$$ the b's cancel

The benchmark: perfect competition

Under perfect competition, price equals marginal cost.

$$
\begin{aligned}
a - bQ &= c && \text{set } P = \text{MC} \\
Q_{\text{comp}} &= \frac{a-c}{b} && \text{solve for } Q \\
P_{\text{comp}} &= c
\end{aligned}
$$

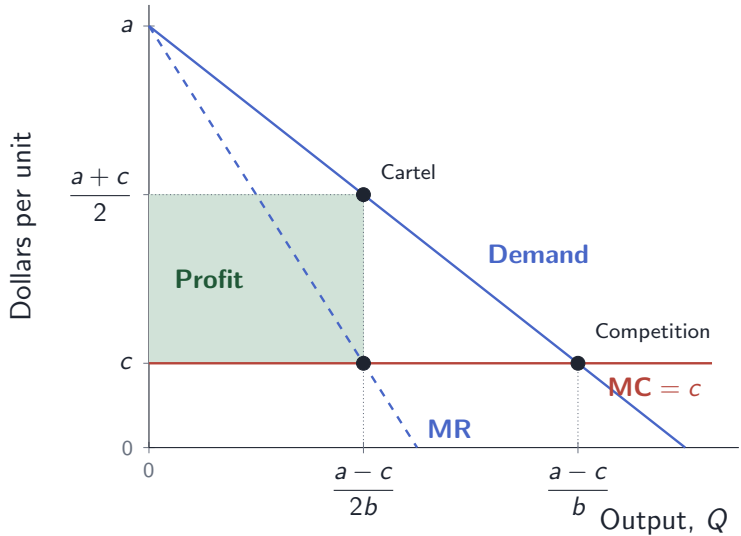
Compare the two outputs:

$$Q_{\text{cartel}} = \frac{a-c}{2b} = \frac{1}{2} \times \frac{a-c}{b} = \frac{1}{2} Q_{\text{comp}}.$$

Finding

A cartel produces half the competitive output and charges $\frac{a+c}{2}$, above marginal cost c.

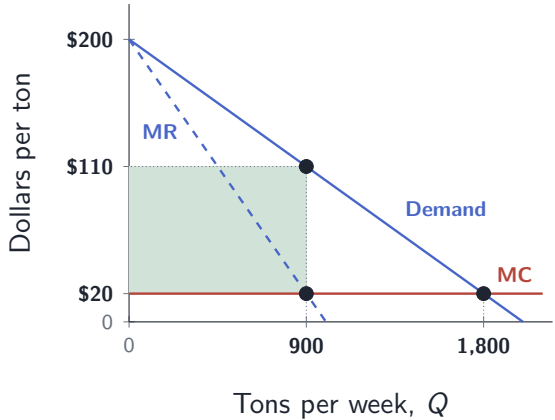
The cartel on the graph



Example: a cartel of two cement firms

$P = 200 - Q/10$, so $a = 200$, $b = 1/10$; $c = \$20$.

1. $Q_{\text{cartel}} = \dfrac{200 - 20}{2 \times 1/10} = 900$ tons.
2. $P_{\text{cartel}} = \dfrac{200 + 20}{2} = \$110$.
3. Profit $= (110 - 20) \times 900 = \$81{,}000$.
4. Split equally: 450 tons and \$40,500 each.
5. Competition: $Q = \dfrac{180}{1/10} = 1{,}800$ tons at \$20.



Part 2 summary

- Collusion: firms agree on quantities or prices.
- A cartel acts as one monopolist: MR = MC.
- $Q_{\text{cartel}} = \dfrac{a-c}{2b}$ and $P_{\text{cartel}} = \dfrac{a+c}{2}$.
- The members split the output and the profit.
- Compared with perfect competition: half the output, and a price above MC.

Check: Part 2

If duopoly firms that are not colluding were able to successfully collude, then

A price and quantity would rise.

B price and quantity would fall.

C price would rise and quantity would fall.

D price would fall and quantity would rise.

Check: Part 2 – answer

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Finding

A cartel restricts output to the monopoly level, so quantity falls and price rises.

Part 3

Why Collusion Breaks Down

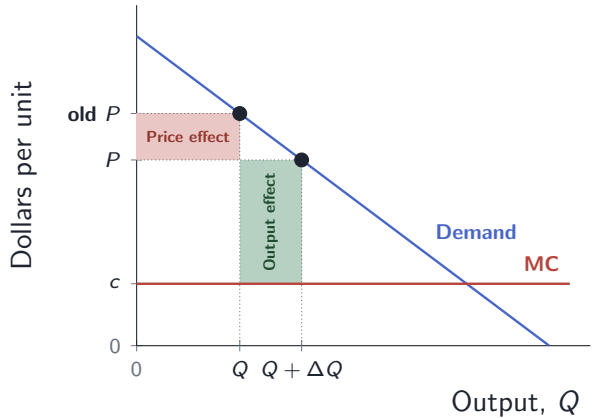
Step 1–2: what one more unit does to joint profit

Joint profit: $\pi = (P - c) \times Q$.

Output rises by ΔQ; price changes by $\Delta P < 0$.

- **Output effect:** each extra unit earns $P - c$: $(P - c)\,\Delta Q > 0$.
- **Price effect:** the Q units already sold fetch ΔP less: $Q\,\Delta P < 0$.

$$\Delta\pi = (P - c)\,\Delta Q + Q\,\Delta P.$$



Step 3: the cartel's choice

- The cartel adds output while one more unit raises joint profit.
- It stops where one more unit ($\Delta Q = 1$) adds nothing:

$$(P - c)\,\Delta Q + Q\,\Delta P = 0.$$

- Divide both sides by ΔQ:

$$P + Q \times \frac{\Delta P}{\Delta Q} = c.$$

- The left side is MR (Chapter 16). The right side is MC. So this is MR = MC.

Finding

At the cartel output, the output effect and the price effect on joint profit exactly cancel.

Step 4: split the price effect between the two firms

Start from the change in joint profit and write $Q = q_A + q_B$:

$$\begin{aligned}
\Delta\pi &= (P - c)\,\Delta Q + Q\,\Delta P && \text{Steps 1–2} \\
&= (P - c)\,\Delta Q + (q_A + q_B)\,\Delta P && \text{replace } Q \text{ with } q_A + q_B \\
&= (P - c)\,\Delta Q + q_A\,\Delta P + q_B\,\Delta P && \text{multiply out the bracket}
\end{aligned}$$

The price effect is now in two pieces: $q_A\,\Delta P$ falls on A's units and $q_B\,\Delta P$ falls on B's units.

Step 5: firm A adds the output on its own

Firm A raises its own output by ΔQ. Firm A's profit counts only firm A's units.

- It gets the whole output effect: $(P - c)\,\Delta Q$.
- It bears the price cut only on its own units: $q_A\,\Delta P$.

$$\Delta \pi_A = (P - c)\,\Delta Q + q_A\,\Delta P.$$

Compare with the last line of Step 4: $\Delta \pi_A$ is $\Delta \pi$ without the last term.

$$\Delta \pi_A = \Delta \pi - q_B\,\Delta P.$$

Step 6: start from the cartel output

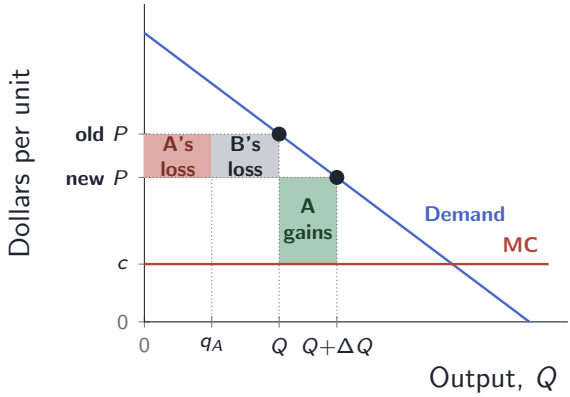
- At the cartel output, one more unit leaves joint profit unchanged (Step 3): $\Delta\pi = 0$.
- Put $\Delta\pi = 0$ into Step 5:

$$\Delta\pi_A = 0 - q_B\,\Delta P = -\,q_B\,\Delta P.$$

- The cartel as a whole gains nothing from the extra unit.
- Firm A gains exactly the loss that the lower price imposes on firm B's units.

Step 7: read the sign

- $\Delta P < 0$, so $-q_B \Delta P > 0$: firm A gains.
- A collects the whole output effect (green).
- A loses only the price cut on its own units (red).
- The price cut on B's units (grey) is B's loss. A ignores it.



Finding

At the cartel output, each member gains by producing more. That is why cartels break down.

Example: one firm cheats

Cement cartel: $P = 200 - Q/10$, $c = \$20$, each firm agreed to 450 tons.

- A raises output to 600 tons; B keeps to 450. $Q = 1{,}050$, so $P = 200 - 105 = \$95$.
- Price falls by \$15: $\Delta P = -15$. A's extra output: $\Delta Q = 150$.
- Output effect: $(95 - 20) \times 150 = +\$11{,}250$.
- Price effect on A's own 450 tons: $450 \times (-15) = -\$6{,}750$.
- A's gain: $11{,}250 - 6{,}750 = +\$4{,}500$. A now earns $40{,}500 + 4{,}500 = \$45{,}000$.
- Price effect on B's 450 tons: $450 \times (-15) = -\$6{,}750$. B now earns \$33,750.
- Joint profit: $78{,}750 - 81{,}000 = -\$2{,}250$. Check Step 5: $\Delta\pi_A = -2{,}250 + 6{,}750 = +\$4{,}500$.

Finding

A 150-ton jump lowers joint profit a little, yet cheating still pays firm A \$4,500.

The payoff table

Each firm chooses 450 tons (keep the agreement) or 600 tons (cheat). Profits are (A, B).

	B makes 450	B makes 600
A makes 450	($40,500, $40,500)	($33,750, $45,000)
A makes 600	($45,000, $33,750)	($36,000, $36,000)

- Both make 600: $Q = 1{,}200$, $P = \$80$, profit $(80 - 20) \times 600 = \$36{,}000$ each.
- If B makes 450, A earns more at 600: $45,000 against $40,500.
- If B makes 600, A earns more at 600: $36,000 against $33,750.

Definition: Dominant strategy

The best choice for a player whatever the other player does. Here: 600 tons, for both firms.

Nash equilibrium and the prisoners' dilemma

Definition: Nash equilibrium

A pair of choices in which each player's choice is its best, given the other's choice. Neither gains by changing its own choice alone.

- At (600, 600), A cannot do better by switching to 450: \$33,750 $<$ \$36,000. Nor can B.
- So (600, 600) is the Nash equilibrium.
- Both would earn more keeping the agreement: \$40,500 each.

Definition: Prisoners' dilemma

A game in which each player's own best choice leaves both worse off than if they cooperated.

Where do the firms end up?

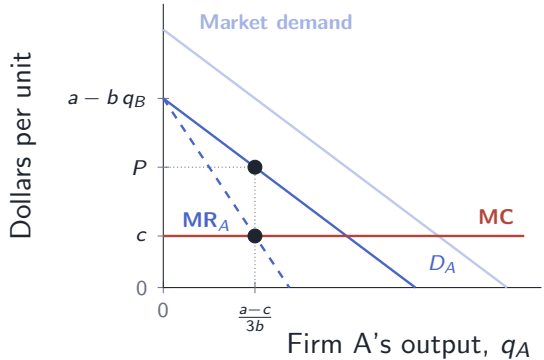
Firm A takes B's output q_B as given. Its demand is D_A.

$$P = (a - b\,q_B) - b\,q_A$$
$$\text{MR}_A = (a - b\,q_B) - 2b\,q_A$$
$$c = a - b\,q_B - 2b\,q_A$$

Equal outputs, $q_A = q_B = q$: $a - 3bq = c$, so

$$q = \frac{a-c}{3b}, \qquad Q = \frac{2(a-c)}{3b}.$$

Cement: $q = \dfrac{180}{3/10} = 600$ tons each.



Part 3 summary

- One more unit: output effect $(P - c)\,\Delta Q$ and price effect $Q\,\Delta P$.
- At the cartel output the two effects on joint profit cancel.
- A single firm bears only its own share of the price effect: $\Delta \pi_A = -\,q_B\,\Delta P > 0$.
- So each member gains by cheating, and the cartel breaks down.
- Dominant strategy: produce more. Nash equilibrium: both produce more, $q = \frac{a-c}{3b}$ each.
- Prisoners' dilemma: both end up with less profit than under the cartel.

Check: Part 3

Cartels are difficult to maintain because

A the monopoly output is very difficult to determine.

B the number of firms is always large.

C costs to the firms in a cartel are continually rising.

D each firm has an incentive to deviate from its agreed output level.

Check: Part 3 – answer

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Finding

At the cartel output, $\Delta\pi_A = -q_B\,\Delta P > 0$: each member gains by producing more.

Part 4

Market Outcomes Compared

Three outcomes from the same demand and cost

Structure	Output	Price	Total profit	P and MC
Monopoly or cartel	$\frac{a-c}{2b}$	$\frac{a+c}{2}$	Highest	$P > \text{MC}$
Oligopoly, two firms	$\frac{2(a-c)}{3b}$	$\frac{a+2c}{3}$	Lower	$P > \text{MC}$
Perfect competition	$\frac{a-c}{b}$	c	Zero	$P = \text{MC}$

The oligopoly price, from demand: $P = a - b \times \frac{2(a-c)}{3b} = a - \frac{2(a-c)}{3} = \frac{a+2c}{3}$.

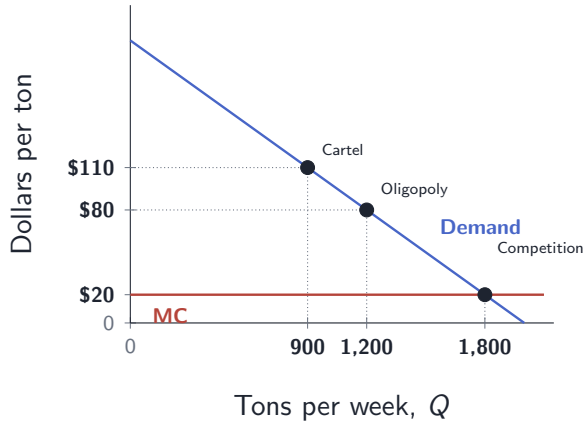
Finding

Oligopoly output lies between monopoly and perfect competition, and so does its price.

The cement market: three outcomes

	Q	P
Cartel	900	\$110
Oligopoly	1,200	\$80
Competition	1,800	\$20

Total profit: \$81,000, then
$2 \times 36{,}000 = \$72{,}000$, then \$0.



All four market structures in the long run

	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Price and MC	$P = \text{MC}$	$P > \text{MC}$	$P > \text{MC}$	$P > \text{MC}$
Long-run profit	Zero	Zero	Can be positive	Can be positive
Why	Free entry	Free entry	Entry is difficult	Entry is blocked
Output	Efficient	Below efficient scale	Below competitive	Lowest

Finding

Only perfect competition sets $P = \text{MC}$. In the other three, some buyers who value a unit above its cost do not get it.

Collusion and the law

- Collusion raises price and cuts output, so buyers lose.
- **Sherman Antitrust Act (1890):** agreements among competitors to fix prices or restrict output are illegal.
- **Clayton Act (1914):** individuals harmed by illegal agreements can sue to recover damages.
- **Example:** in 1996 Archer Daniels Midland agreed to plead guilty to fixing the prices of lysine and citric acid, two food and feed additives, and paid a $100 million fine.

Source: U.S. Department of Justice, press release, 15 October 1996.

Part 4 summary

- Cartel: output $\frac{a-c}{2b}$, the lowest; price $\frac{a+c}{2}$, the highest.
- Oligopoly with two firms: output $\frac{2(a-c)}{3b}$; price $\frac{a+2c}{3}$.
- Perfect competition: output $\frac{a-c}{b}$; price c.
- Oligopoly lies between monopoly and perfect competition.
- Price-fixing is illegal under the Sherman Act.

Check: Part 4

The equilibrium quantity in markets characterized by oligopoly is

A higher than in monopoly markets and higher than in perfectly competitive markets.

B higher than in monopoly markets and lower than in perfectly competitive markets.

C lower than in monopoly markets and higher than in perfectly competitive markets.

D lower than in monopoly markets and lower than in perfectly competitive markets.

Check: Part 4 – answer

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C lower than in monopoly markets and higher than in perfectly competitive markets.

D lower than in monopoly markets and lower than in perfectly competitive markets.

Finding

$\frac{a-c}{2b} < \frac{2(a-c)}{3b} < \frac{a-c}{b}$. In cement: $900 < 1{,}200 < 1{,}800$.

Which rule answers which question

Question	Rule
What defines an oligopoly?	A few sellers, similar products, difficult entry, interdependence.
What does a cartel produce?	The monopoly output: MR = MC, $Q = \frac{a-c}{2b}$.
Why does a cartel break down?	Each member gains by producing more: $\Delta\pi_A = -\,q_B\,\Delta P > 0$.
What is a dominant strategy?	The best choice whatever the rival does.
What is a Nash equilibrium?	Each choice is best given the other's choice.
Where does oligopoly output lie?	Between monopoly and perfect competition.
Is collusion legal?	No: price-fixing is illegal under the Sherman Act.